Multivariate vs. Univariate Forecasting of U.S. Wholesale Electricity Prices

Group #12: Model Citizens
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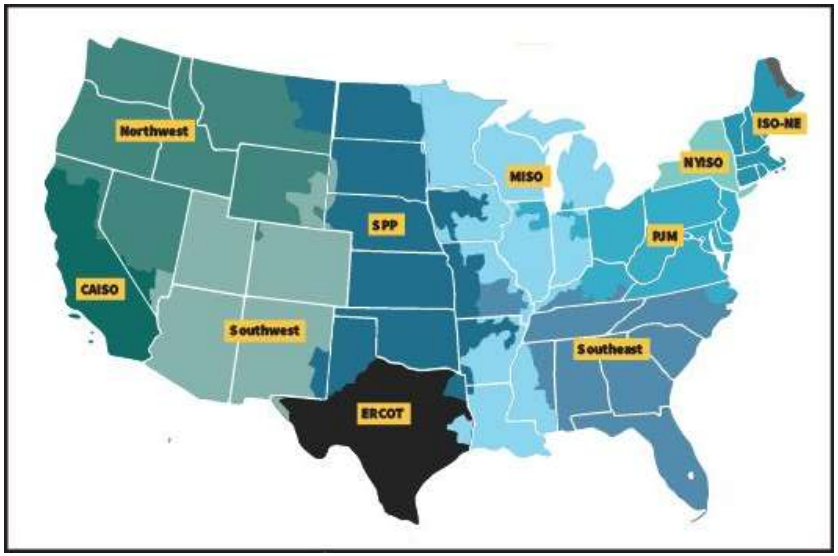
Background & Motivation : Why does this matter?

- Wholesale electricity prices shape how electricity is generated, transmitted, and priced, though most consumers don't see them directly.
- Prices depend on fuel costs, plant operations, transmission, weather, and location, creating complexity and volatility.
- Understanding cross-hub relationships and accurate forecasts is critical for energy producers, market operators, and policymakers.



Research Question: Do multivariate models forecast regional wholesale electricity prices more accurately than univariate models?

Hypothesis & Modeling Approach

Multivariate models that use cross-hub dependencies produce more accurate forecasts of U.S. wholesale electricity prices than univariate hub-by-hub models.

Modeling Approach:

- Fit univariate SARIMA models per hub.
- Build multivariate VAR model across hubs.
- Create 1-, 3-, 6-, and 12-month forecasts.
- Refit best model to generate 2027 forecasts and compare to the U.S. Energy Information Administration's Short-Term Energy Outlook predictions

Data Acquisition & Explanation

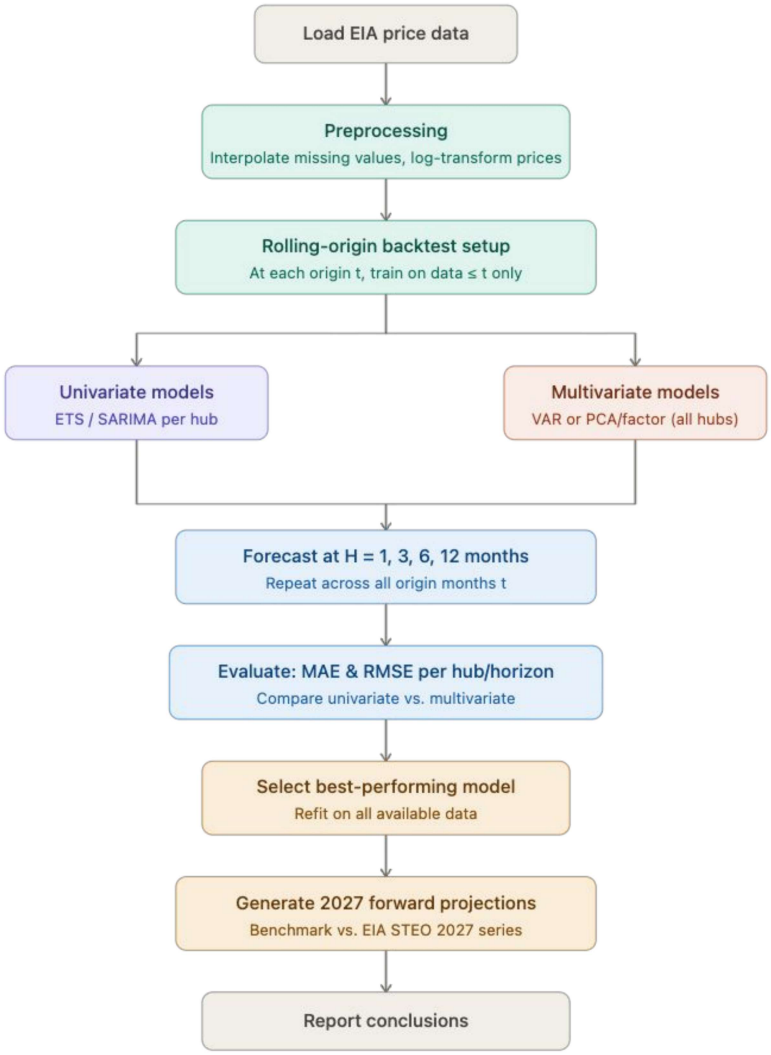
Source: Monthly wholesale electricity prices from the U.S. Energy Information Administration, compiled via Fetch Series

Contains prices (USD/MWh) for 11 regional U.S. hubs from 2010 through 2027, including ERCOT North, ISO-NE Internal, NYISO Hudson Valley, PJM Western, etc.

- Tabular time series data with 12 columns (Date + 11 hubs) covering 204 months

Publicly available dataset, contains no sensitive information

Column	Description	Potential Responses
Date String	The end of each month from 2010-2027.	2027-12-31, 2025-9-30
US - ERCOT North hub	Wholesale electricity price (USD / MWh) in the United States (ERCOT North hub)	34.45978261, 47.5943
US - CAISO SP15 zone	Wholesale electricity price (USD / MWh) in the United States (CAISO SP15 zone)	50.66786172, 38.40781166
US - ISO-NE Internal hub	Wholesale electricity price (USD / MWh) in the United States (ISO-NE Internal Hub)	67.12396875, 50.70175
US - NYISO Hudson Valley zone	Wholesale electricity price (USD / MWh) in the United States (NYISO Hudson Valley Zone)	71.001875, 82.7533631
US - PJM Western hub	Wholesale electricity price (USD / MWh) in the United States (PJM Western Hub)	79.27900158, 64.09666463
US - Midcontinent ISO Illinois hub	Wholesale electricity price (USD / MWh) in the United States (Midcontinent ISO Illinois hub)	49.84491477, 36.47159375
US - SPP ISO South hub	Wholesale electricity price (USD / MWh) in the United States (SPP ISO South hub)	53.50369048, 48.48042614
US - SERC index Into Southern	Wholesale electricity price (USD / MWh) in the United States (SERC index Into Southern)	40.82142857, 29.175
US - FRCC index Florida Reliability	Wholesale electricity price (USD / MWh) in the United States (FRCC index Florida Reliability)	30.625, 28.97727273
US - Northwest index Mid-Columbia	Wholesale electricity price (USD / MWh) in the United States (Northwest index Mid-Columbia)	19.24318182, 27.957
US - Southwest index Palo Verde	Wholesale electricity price (USD / MWh) in the United States (Southwest index Palo Verde)	24.4855, 46.59954546


Load EIA price data
Preprocessing
Interpolate missing values, log-transform prices
Rolling-origin backtest setup
At each origin t, train on data ≤ t only
Univariate models
ETS / SARIMA per hub
Multivariate models
VAR or PCA/factor (all hubs)
Forecast at H = 1, 3, 6, 12 months
Repeat across all origin months t
Evaluate: MAE & RMSE per hub/horizon
Compare univariate vs. multivariate
Select best-performing model
Refit on all available data
Generate 2027 forward projections
Benchmark vs. EIA STEO 2027 series
Report conclusions

Tricky Analysis Decision

Decision: Backtesting approach that predicts future prices month by month while evaluating multiple forecast horizons (1, 3, 6, and 12 months).

Why it was tricky:

- Needed to ensure no data leakage from future months.
- Had to choose which horizons (1, 3, 6, 12 months) best represented both short- and long-term forecasting.
- Decided on MAE and RMSE as error metrics to fairly compare univariate and multivariate models.

Impact:

Affected out-of-sample accuracy evaluation and final model selection.

Determined the reliability of 2027 forecasts against EIA STEO projections.

Biases & Uncertainty Validation

1. Missing Data & Interpolation

- ERCOT_North: Jan–Nov 2010 missing
- SPP_South: Jan 2010–Feb 2014 missing
- Fix: Linear time-based interpolation
- Uncertainty: Counted missing values before/after interpolation; interpolated points flagged. Variance in interpolated periods is likely underestimated.

2. Skewed Distributions & Variance

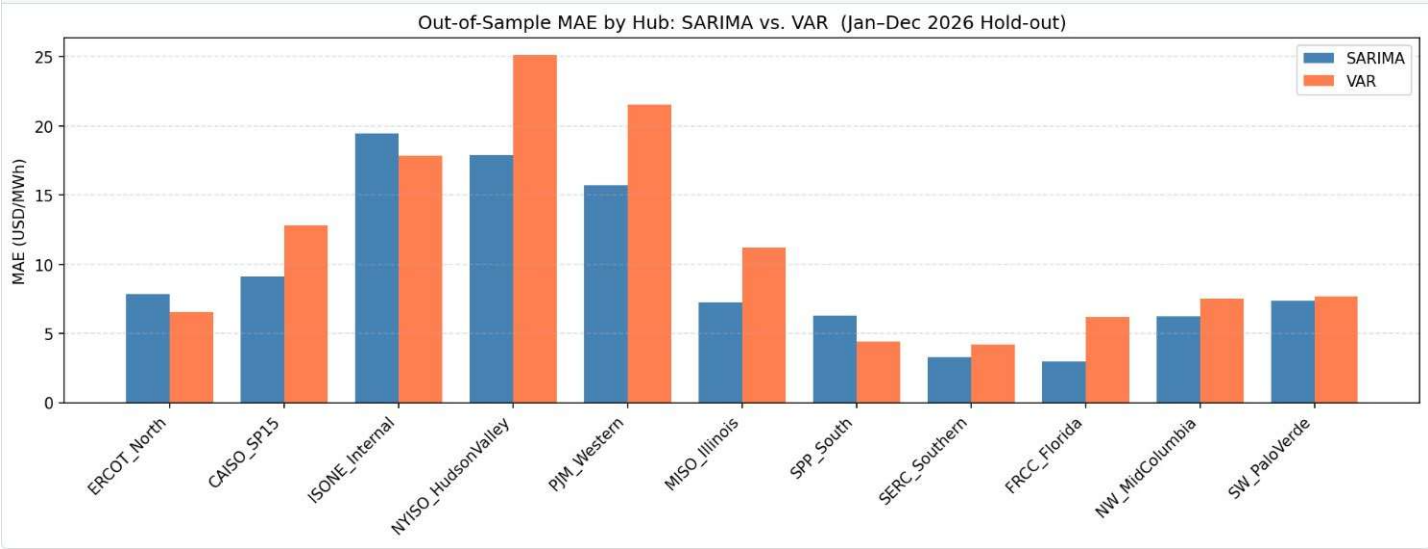
- Prices are right-skewed; occasional near-zero/negative values
- Fix: Log transform (clip at 0.01)
- Uncertainty: Compared pre- and post-transform descriptive statistics; log transformation stabilizes variance, reducing the impact of extreme outliers. Forecast errors are quantified via MAE/RMSE in hold-out data.

3. Limited Hub Histories

- Some hubs have shorter historical records → higher forecast uncertainty
- Fix: VAR leverages correlations across hubs
- Uncertainty: Hubs with less data showed larger MAE/RMSE; cross-hub correlations in VAR reduce but do not eliminate uncertainty.

Results & Conclusion

VAR, which uses multiple hubs together, did help improve forecasts for a few hubs like ERCOT North, ISONE Internal, and SPP South, but overall SARIMA had lower mean absolute error, meaning its predictions were in many cases more accurate.



Next Steps

1. Exploration

- Explore hybrid models combining univariate and multivariate approaches
- Analyze external factors (weather, demand, fuel prices) to enhance forecasts

2. Improvements

- Test more advanced models (machine learning or factor models)
- Improve handling of missing data beyond interpolation

3. Questions

- Why do some hubs show consistently higher forecast error?
- Can combining univariate and multivariate models improve overall forecasts?

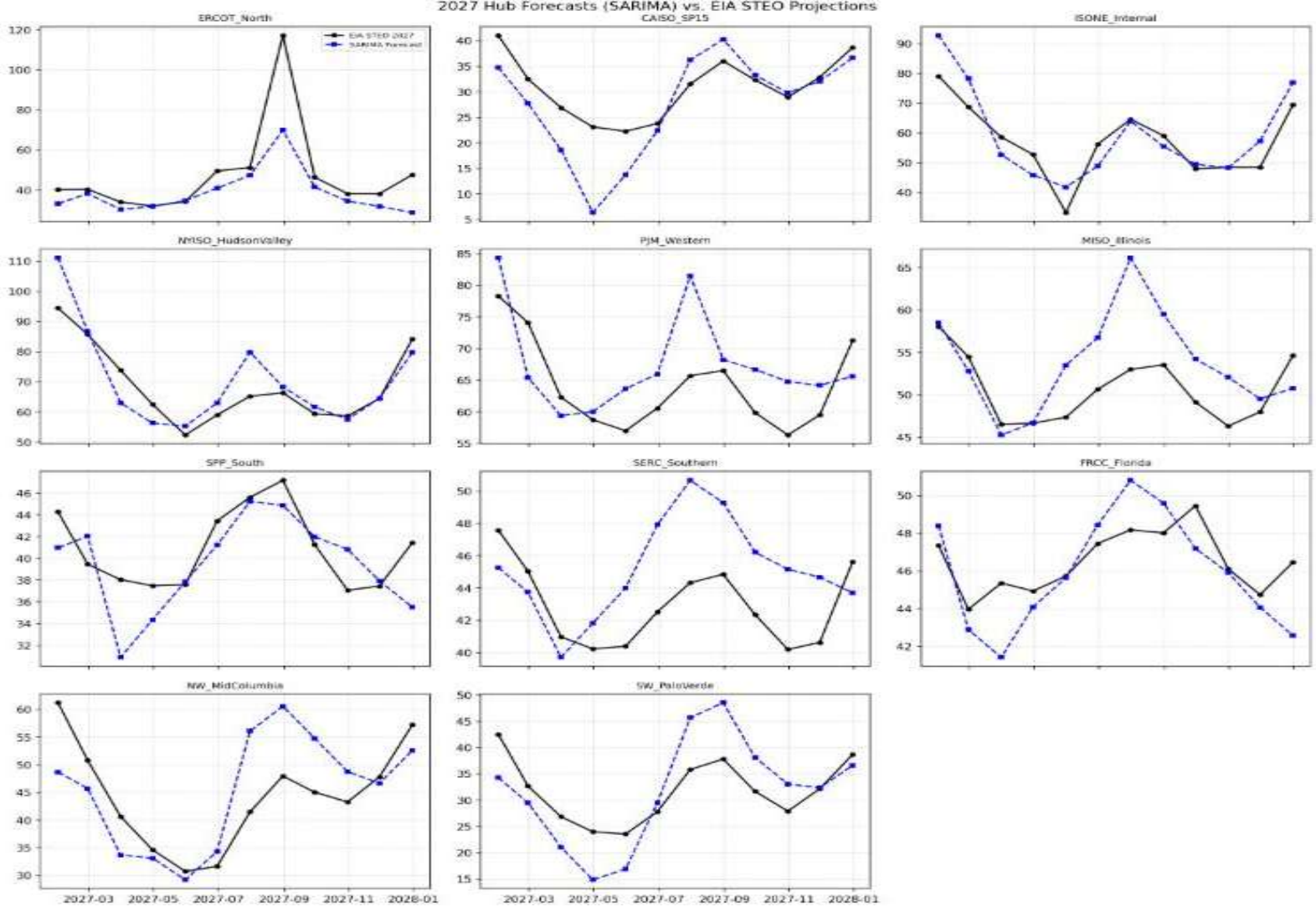
References

[1] shainabanduri, “Project-1-DS4002,” GitHub repository. Available: https://github.com/shainabanduri/Project-1-DS4002/tree/main (accessed March 22, 2026).

[2] “US Wholesale electricity prices (monthly) - FetchSeries Time Series Download,” *Fetchseries.com*, 2026. https://www.fetchseries.com/electricity/us-wholesale-electrictiy-prices-monthly/(accessed March 22, 2026).

[3] U.S. Energy Information Administration, “Prices and factors affecting prices - U.S. Energy Information Administration (EIA),” Eia.gov, 2016. https://www.eia.gov/energyexplained/electricity/prices-and-factors-affecting-prices.php (accessed March 22, 2026)

[4]A. Ganczarek-Gamrot, A. Gorczyca-Goraj, K. Pilot, and K. Kania, “Electricity Price Volatility and the Performance of Machine Learning Forecasting Models in European Energy Markets,” Energies, vol. 18, no. 24, p. 6535, Dec. 2025, doi: https://doi.org/10.3390/en18246535. (accessed March 22, 2026)


2027 Hub Forecasts (SARIMA) vs. EIA STEO Projections
ERCOT_North
EIA STEO 2027
SARIMA Forecast
CAISO_SP15
ISONE_Internal
NYISO_HudsonValley
PJM_Western
MISO_Illinois
SPP_South
SERC_Southern
FRCC_Florida
NW_MidColumbia
SW_PaloVerde
2027-03
2027-05
2027-07
2027-09
2027-11
2028-01